

AWS Lambda Interview Questions and Answers

A concise set of interview questions on AWS Lambda covering triggers, cold starts, concurrency, layers, execution roles, and versioning concepts frequently asked in interviews.

Q1. What is AWS Lambda?

AWS Lambda is a serverless compute service that runs code in response to events without requiring you to provision or manage servers. You pay only for the compute time consumed while your function is executing.

Q2. What triggers can invoke a Lambda function?

Lambda functions can be invoked by many event sources, including API Gateway requests, S3 object events, DynamoDB streams, SQS messages, SNS notifications, EventBridge rules, and Kinesis streams, among others.

Q3. What is a cold start in Lambda?

A cold start happens when Lambda needs to initialize a new execution environment for a function, which includes downloading the code, starting the runtime, and running any initialization code outside the handler. This adds latency compared to a warm invocation that reuses an existing environment.

Q4. How can cold start latency be reduced?

Common approaches include using Provisioned Concurrency to keep environments pre-initialized, minimizing package size and dependencies, choosing a faster-starting runtime, and moving heavy initialization logic outside the per-invocation handler code so it runs once per environment.

Q5. What is the difference between Lambda concurrency and provisioned concurrency?

Concurrency refers to the number of function instances running at the same time to handle simultaneous requests. Provisioned concurrency pre-initializes a specified number of execution environments so they are ready to respond immediately, avoiding cold starts for that reserved capacity.

Q6. What are Lambda layers?

Layers are packaged sets of libraries, custom runtimes, or other dependencies that can be attached to one or more functions. They let you share common code across functions without bundling it into each function's own deployment package.

Q7. What is an execution role in Lambda?

An execution role is an IAM role that grants a Lambda function permission to access other AWS services and resources, such as writing logs to CloudWatch or reading from an S3 bucket. Lambda assumes this role automatically when the function runs.

Q8. What are Lambda versions and aliases used for?

Versions capture an immutable snapshot of a function's code and configuration at a point in time. Aliases are pointers to a specific version, which makes it possible to route traffic between versions (for example, for gradual rollouts) without changing the code that invokes the function.

Q9. What is the maximum execution timeout for a Lambda function?

A Lambda function can run for up to 15 minutes per invocation. Workloads that need to run longer than this are generally better suited to services like AWS Fargate, EC2, or Step Functions orchestrating multiple shorter steps.